

Netflix Movies & TV Shows Dataset – Summary Report

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Project: Data Cleaning and Exploratory Data Analysis (EDA)

Dataset: Netflix Movies and TV Shows Dataset (Kaggle)

Project Overview

The objective of this project was to clean, validate, explore, and visualize the Netflix Movies and TV Shows dataset. The dataset was prepared by identifying and resolving data quality issues before performing exploratory data analysis to uncover meaningful patterns and insights.

Cleaning Challenges Encountered

Several data quality issues were identified during the cleaning process:

- Missing values were found in the `director`, `cast`, `country`, `date_added`, `rating`, and `duration` columns.
- Some records contained leading spaces in the `date_added` column, which caused errors when converting the column to a datetime format.
- The `rating` column contained incorrect values (74 min, 84 min, and 66 min) that belonged in the `duration` column.
- Missing categorical values were handled by replacing them with "Unknown", while rows with missing `date_added` values were removed because they represented a very small proportion of the dataset.
- Duplicate records were checked, and none were found.

Key Findings

The exploratory data analysis revealed several important insights:

- Movies make up a larger proportion of Netflix's content library than TV Shows.
- Netflix significantly expanded its content catalog over time, with content additions increasing in later years.
- A small number of countries contribute the majority of Netflix's available content.
- Content rated for mature audiences (such as TV-MA and TV-14) appears more frequently than other ratings.
- Certain genres occur more frequently than others, indicating Netflix's emphasis on popular content categories.

Conclusion

The dataset was successfully cleaned, standardized, validated, and analyzed. The cleaning process improved the quality and reliability of the data, while the exploratory analysis and visualizations provided valuable insights into Netflix's content library. The final dataset is suitable for further analysis and supports informed, data-driven decision-making.